

7. Integration Techniques

7.8. Improper Integrals

7.8a Learning Objectives

- Identify type 1 and type 2 improper integrals
- Rewrite improper integrals as limits of proper integrals
- Use the comparison test to determine whether an integral converges or diverges

7.8.1. Motivation

So far we have worked with **proper** definite integrals, meaning

1. the domain of integration is finite, $[a, b]$, and
2. the range of the integrand is finite on the domain

What if we break these rules? $\int_{-\infty}^{\infty} \frac{1}{x} dx$

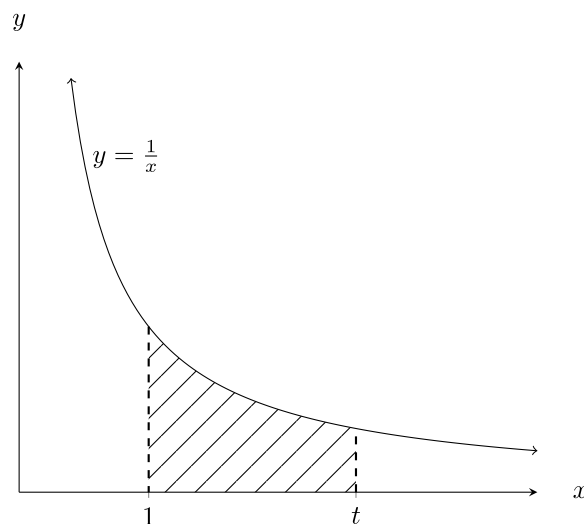
has infinite bounds and a vertical asymptote.

Remark

Why would we want to break these rules? As you will see in 8.5, there are some very important applications of improper integrals in the realm of probability and statistics. We will get there in a little, let's build up the tools to evaluate these first.

How might we find the area under the curve $f(x) = \frac{1}{x}$ on $[1, \infty)$?

With the current methods we have, we can evaluate $\int_1^t \frac{1}{x} dx$ for any finite-valued $t > 1$.



Then as we let $t \rightarrow \infty$, the true area under the curve is more accurately approximated. Thus, we can take the limit of these integrals only covering a portion of the region.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx$$

7.8.2. Type 1 Improper Integrals

Definition 7.8.1 Type 1 Improper Integrals

These are integrals with infinite limits:

1. If $f(x)$ is continuous on (a, ∞)

$$\int_a^{\infty} f(x) \, dx = \lim_{b \rightarrow \infty} \int_a^b f(x) \, dx$$

2. If $f(x)$ is continuous on $(-\infty, b)$

$$\int_{-\infty}^b f(x) \, dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) \, dx$$

3. If $f(x)$ is continuous on $(-\infty, \infty)$

$$\int_{-\infty}^{\infty} f(x) \, dx = \int_{-\infty}^c f(x) \, dx + \int_c^{\infty} f(x) \, dx \quad \text{for any real number } c$$

Note

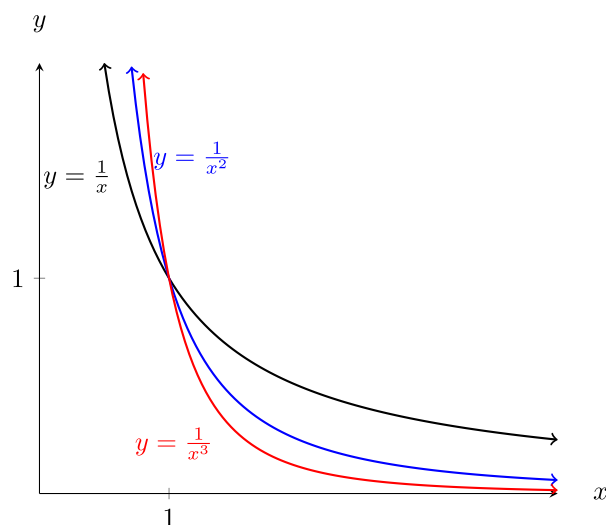
If this limit exists and is a finite number, we say the improper integral *converges* and the limit is the value of the improper integral. Otherwise, we say the integral *diverges*.

Example 7.8.1

For what values of p does the integral $\int_1^{\infty} \frac{1}{x^p} \, dx$ converge? What is the value when it converges?

Solution

Notice that for different values of p that $y = \frac{1}{x^p}$ approaches $y = 0$ at a different rate. This means that for some values of p , the function will decrease fast enough to converge, but not always.



- If $p = 1$:

$$\int_1^{\infty} \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx = \lim_{t \rightarrow \infty} [\ln|x|]_1^t = \lim_{t \rightarrow \infty} \ln(t) = \infty$$

So the integral diverges in this case.

- If $p \neq 1$:

$$\begin{aligned} \int_1^{\infty} \frac{1}{x^p} dx &= \lim_{t \rightarrow \infty} \int_1^t x^{-p} dx \\ &= \lim_{t \rightarrow \infty} \left[\frac{1}{-p+1} x^{-p+1} \right]_1^t \\ &= \lim_{t \rightarrow \infty} \left[\frac{1}{-p+1} \frac{1}{t^{p-1}} - \frac{1}{-p+1} \right] \end{aligned}$$

This last limit converges if $p > 1$ and diverges if $p < 1$. If $p > 1$, it converges to $\frac{-1}{1-p} = \frac{1}{p-1}$

Example 7.8.2

Find the area under the curve of $y = \frac{\ln(x)}{x^2}$ for $x \geq 1$.

Solution

We first rewrite the improper integral as a limit of a proper integral,

$$\int_1^{\infty} \frac{\ln(x)}{x^2} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{\ln(x)}{x^2} dx$$

Now we integrate $\int_1^b \frac{\ln(x)}{x^2} dx$ with our normal methods, just writing $\lim_{x \rightarrow b} [\]$ around everything...

$$\int_1^{\infty} \frac{\ln(x)}{x^2} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{\ln(x)}{x^2} dx$$

we need integration by parts for this integral, so

$$\begin{aligned} u &= \ln(x), \quad dv = \frac{1}{x^2} dx, \\ du &= \frac{1}{x} dx, \quad v = \int \frac{1}{x^2} dx = -\frac{1}{x} \end{aligned}$$

then

$$\begin{aligned}
\int_1^{\infty} \frac{\ln(x)}{x^2} dx &= \lim_{b \rightarrow \infty} \int_1^b \frac{\ln(x)}{x^2} dx \\
&= \lim_{b \rightarrow \infty} \left[\ln(x) \left(-\frac{1}{x} \right) \Big|_1^b - \int_1^b \left(-\frac{1}{x} \right) \frac{1}{x} dx \right] \\
&= \lim_{b \rightarrow \infty} \left[\frac{-\ln(x)}{x} - \frac{1}{x} \right] \Big|_1^b \\
&= \lim_{b \rightarrow \infty} \left[\frac{-\ln(b)}{b} - \frac{1}{b} + \frac{\ln(1)}{1} + \frac{1}{1} \right] \\
&= \lim_{b \rightarrow \infty} \left[\frac{-\ln(b)}{b} - \frac{1}{b} + 1 \right] \\
&= \lim_{b \rightarrow \infty} \frac{-\ln(b)}{b} + \lim_{b \rightarrow \infty} 1 - \frac{1}{b} \\
&\stackrel{\text{LH}}{=} \lim_{b \rightarrow \infty} \frac{-\frac{1}{b}}{1} + 1 \\
&= 0 + 1 = 1
\end{aligned}$$

Warning

Recall that ∞ is not a number. We cannot evaluate things at infinity, we have to discuss these things in terms of limits. Notation is going to play a large role in this section, so pay close attention to it. We always rewrite these in terms of limits.

Example 7.8.3

Evaluate $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$.

Solution

We can split this integral up at any real number, but thinking ahead the antiderivative of this is $\arctan(x)$ and $\arctan(0) = 0$ so we will split it at 0. It is not necessary that you think ahead in this way, but it can be nice.

$$\begin{aligned}
\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx &= \int_{-\infty}^0 \frac{1}{1+x^2} dx + \int_0^{\infty} \frac{1}{1+x^2} dx \\
&= \lim_{t \rightarrow -\infty} \int_t^0 \frac{1}{1+x^2} dx + \lim_{r \rightarrow \infty} \int_0^r \frac{1}{1+x^2} dx \\
&= \lim_{t \rightarrow -\infty} \arctan(x) \Big|_t^0 + \lim_{r \rightarrow \infty} \arctan(x) \Big|_0^r \\
&= \lim_{t \rightarrow -\infty} -\arctan(t) + \lim_{r \rightarrow \infty} \arctan(r) \\
&= \frac{\pi}{2} + \frac{\pi}{2} \\
&= \pi
\end{aligned}$$

7.8a Section Summary: